

Problem Statement

Clinical Validation of AI based Diagnostic Screening System for KUB Risk detection and Classification Using X-Ray

KUB (Kidney, Ureter, and Bladder) X-ray imaging, typically performed as an AP (Anteroposterior) supine view, is a primary diagnostic, low-cost, and widely available tool used to detect and classify urinary tract abnormalities. While non-contrast CT (NCCT) is the gold standard for renal calculi, KUB X-rays are critical for assessing stone burden, classification, and tracking, especially with emerging AI-enhanced detection techniques that can achieve high accuracy in identifying kidney stones.

Here is a breakdown to help you structure your project:

1. Primary Parameters and Symptoms for Training

To build a "clinical-grade" model, you need to look beyond just the pixels. Consider these two categories:

A. Radiographic Features (Image Parameters):

- **Radiopacity:** The "brightness" of a potential stone. Calcium-based stones are highly radiopaque, while uric acid stones are radiolucent (invisible on X-ray).
- **Anatomical Location:** Your model should be "aware" of the **UTJ** (Ureteropelvic Junction) and **UVJ** (Ureterovesical Junction), as these are the most common sites for stone impaction.
- **Size and Shape:** Small stones (< 3 mm) are often missed. Training the model to detect subtle "grain-like" densities is key.
- **Indirect Signs:** Look for a lack of psoas muscle shadow (which can indicate inflammation or fluid) or abnormal bowel gas patterns.

B. Clinical Metadata (Auxiliary Inputs):

If you can feed these into your model's dense layers along with the image features, accuracy will spike:

- **Symptoms:** Renal colic (flank pain), hematuria (blood in urine), and dysuria.
- **Demographics:** Age and Gender (calculi prevalence varies by demographic).
- **History:** Previous history of urolithiasis.

2. Existing Models for Fine-Tuning

You don't need to build from scratch. In medical imaging, **Transfer Learning** is the standard.

- **VGG16 / VGG19:** Heavily used in KUB research due to its simplicity. It has been shown to achieve up to **97% accuracy** for stone vs. normal classification.
- **ResNet-50 / ResNet-101:** Excellent for deeper feature extraction. The "residual" connections help the model identify tiny stones without losing spatial information.
- **EfficientNet (B0 to B7):** If you are targeting a system for clinics with limited hardware, EfficientNet provides high accuracy with much fewer parameters.
- **YOLOv8 or Faster R-CNN:** Use these if your goal is **Object Detection** (drawing a box around the stone) rather than just classification.

3. Suggested Labeled Datasets

Finding high-quality, labeled KUB X-rays is the hardest part of this project. Here are the best places to look:

- **Roboflow Universe:** Search for "KUB" or "Kidney Stones." There are several community-contributed datasets with bounding box annotations specifically for X-rays.
- **Kaggle:** Look for the "Kidney Stone Images" dataset. Note that many are CT-based; however, you can use CT data to create **DRRs (Digitally Reconstructed Radiographs)**—synthetic X-rays—to augment your training set.
- **The Cancer Imaging Archive (TCIA):** While focused on cancer, their abdominal collections often contain thousands of KUB-view images that can be re-labeled.
- **MIMIC-CXR (Lower Slices):** Though it's a chest X-ray dataset, many images include the upper abdomen (kidneys). You can crop these to train the kidney-detection portion of your model.

4. Expected Diagnostic Inference

After inputting a PA or AP X-ray, your "Inference Output" should provide three levels of data to the clinician:

Feature	Type of Output	Clinical Value
Probability Score	Scalar (e.g., 0.89)	Tells the doctor how confident the AI is that a risk exists.
Localization	Heatmap (Grad-CAM) or Bounding Box	Points exactly to where the suspected stone or mass is located.
Classification	Label (e.g., "Ureteric Calculus")	Categorizes the risk (Kidney stone, Bladder stone, or Phlebolith).
Risk Stratification	Severity Score (Low/High)	Based on size and location (e.g., a stone at the UVJ is "High Risk" for obstruction).

Clinical Tip: AP vs. PA View

In KUB imaging, the **AP (Anterior-Posterior)** view is preferred. Why? Because the kidneys are retroperitoneal (located toward the back). In an AP view, the kidneys are closer to the detector, which reduces magnification and provides a much sharper image for your AI to analyze.